

CISC 211 Week 1 HW Boolean Algebra

1) Simplify Boolean expression

$$F = \underline{A'B'C'} + \underline{A'B'C} + \underline{A'BC} + \underline{AB'C} + \underline{ABC'}$$

$$a+(bc) = (a+b)(a+c) \quad a(b+c) = ab+ac$$

ab
and gate

a+b
or gate

$$a+a' = 1 = 1+0$$

$$aa' = 0 = 0+1 = 1+0$$

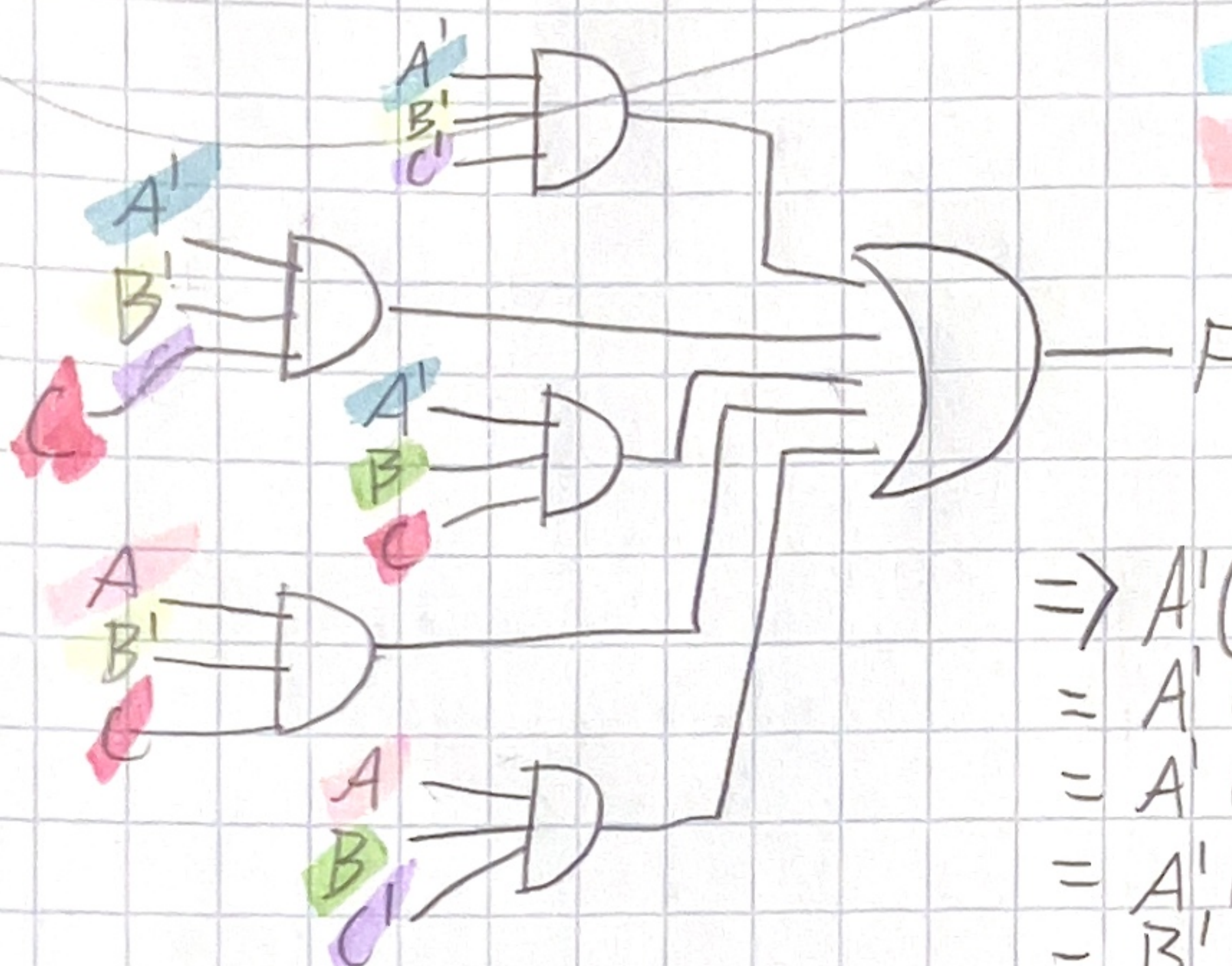
$$y = a+b = y = a+a'b$$

notes

a'
not

a.1

F	BC			
	00	01	11	10
A	0	1	1	0
	1	0	0	1



$$F = A'(B'C' + B'C + BC) + A(B'C + BC')$$

$$\begin{aligned} &\Rightarrow A'(B'(C'+C) + BC) + A(B'C + BC') \\ &= A'(B'(1) + BC) + A(B'C + BC') \\ &= A'(B' + C) + AB'C + ABC' \\ &= A'B' + A'C + AB'C + ABC' \\ &= B'(A' + AC) + A'C + ABC' \\ &= B'(A' + C) + A'C + ABC' \\ &= A'B' + B'C + A'C + ABC' \end{aligned}$$

$$F = C(A' + B') + A'B' + C'(AB)$$

2) Input
m = motion sensed t = test mode

Output

i = illuminate lamp

lamp off if:

m and t'

t and m

t and m'

$$\begin{aligned} &mt' + tm + tm' \\ &= mt' + t(m + m') \\ &= mt' + t \end{aligned}$$